Conformal Coverage Under Quantisation: A Case Study on Edge-Deployed YOLOv8

Faizan Shafi Seroo
School of Physics, Engineering and Computer Science
University of Hertfordshire
Hatfield, United Kingdom
faizanshafi454@gmail.com

Grigorios Skaltsas
School of Physics, Engineering and Computer Science
University of Hertfordshire
Hatfield, United Kingdom
g.skaltsas@herts.ac.uk

***Abstract*—Post-training quantisation is a standard technique for deploying object detectors on CPU-class edge hardware, yet its effect on formal uncertainty guarantees has not been studied. This paper presents the first empirical investigation of split conformal prediction (CP) applied to a YOLOv8n detector exported via Intel OpenVINO at FP32, FP16, and INT8 precision, evaluated on subsets of COCO128 and COCO val2017. We design a 24-condition experiment (3 precisions $\times$ 8 input conditions: clean, three levels of Gaussian blur, three JPEG compression qualities, and additive Gaussian noise) and show that, while the FP32 model achieves an empirical coverage rate of 0.9165 on clean test data, the FP16 and INT8 quantised models fall below the nominal 90% target (0.8929 and 0.8976 respectively). Under degradation, FP32 violates coverage in 1 of 8 conditions, compared to 6/8 for FP16 and 3/8 for INT8. We further show that the convergence of mean nonconformity scores across precisions under degradation mirrors the confidence convergence phenomenon reported in our earlier calibration study, and provide a formal reliability interpretation of how quantisation and input degradation jointly erode conformal guarantees. Our results demonstrate that post-training quantisation should be treated as a model change from the standpoint of conformal prediction: coverage guarantees must be re-established empirically after compression and validated under realistic deployment conditions.**

***Index Terms*—conformal prediction, quantisation, YOLOv8, OpenVINO, edge computing, object detection, uncertainty quantification, coverage guarantee, distribution shift**

I. Introduction

Edge-deployed object detectors are now embedded in autonomous vehicles, urban surveillance infrastructure, and industrial monitoring pipelines, where models must run under strict latency and memory budgets on CPU-class hardware. In this setting, post-training quantisation has become a standard technique for compressing models by reducing numerical precision from 32-bit floating point (FP32) to half-precision (FP16) and 8-bit integer (INT8) formats, yielding substantial throughput gains without retraining [18]–[20]. While a large body of work has demonstrated that carefully calibrated INT8 pipelines can preserve mean Average Precision (mAP) within a few percentage points of FP32 baselines [13], [17], evaluations overwhelmingly focus on accuracy and frames-per-second metrics.

For safety-critical deployments, these metrics are incomplete. Downstream decision-making components rely on detector confidence scores to apply thresholds and business logic; a model that retains mAP but becomes systematically overconfident after quantisation can be more dangerous than a slightly less accurate but honestly calibrated one, because high-confidence false positives and false negatives are more likely to pass unnoticed through monitoring layers [15]. Our previous work characterised this risk through confidence calibration analysis (Expected Calibration Error, reliability diagrams) and discovered a *confidence convergence phenomenon*: a 0.382 confidence gap between FP32 and INT8 models on clean data collapses to under 0.034 under any degraded input condition [1].

Conformal prediction (CP) offers a principled, complementary perspective. CP constructs prediction sets with a user-specified marginal coverage guarantee $1 - \alpha$ under minimal assumptions, typically exchangeability of calibration and test data [2]. Split conformal prediction (SCP) is particularly attractive in practice as it can be applied post-hoc to any black-box model using a held-out calibration set [3], [4]. Recent work has begun applying CP to object detection via two-step conformal schemes that handle class labels and bounding-box uncertainty simultaneously [5]–[7]. Meanwhile, several studies have demonstrated that conformal guarantees can degrade under distribution shift, including image corruption and domain shift [8], [9]. The interaction between *quantisation* and conformal coverage has only recently been studied for large language models, where compression can significantly reduce coverage while leaving accuracy unchanged [10]; to our knowledge, no equivalent study exists for real-time object detectors on edge hardware.

This paper addresses this gap. Building directly on the calibration study in [1], we investigate whether YOLOv8n detectors exported at FP32, FP16, and INT8 via Intel OpenVINO maintain the nominal conformal coverage guarantee, both on clean test images and under the same controlled degradations studied previously.

Contributions. The main contributions of this work are:

- We present, to our knowledge, the first empirical study of split conformal prediction applied to a quantised edge-

deployed object detector (YOLOv8n, FP32/FP16/INT8 via OpenVINO) on COCO data.
- We design a 24-condition evaluation matrix (3 precisions $\times$ 8 input conditions) on a held-out subset of COCO val2017 mirroring the degradation pipeline of [1], including Gaussian blur, JPEG compression, and additive Gaussian noise.
- We show that FP16 and INT8 quantised models fail to meet the nominal 90% coverage guarantee on clean data and violate coverage in 6/8 and 3/8 conditions respectively, compared to 1/8 for FP32, despite having similar calibration statistics.
- We provide a formal reliability interpretation of confidence convergence, connecting Phase-1 findings to conformal nonconformity score distributions and demonstrating that tail behaviour, not mean score alone, determines coverage.

II. Related Work

A. Calibration and Confidence in Quantised Detectors

The calibration of deep neural networks was first systematically studied by Guo et al. [15], who showed that modern classifiers are often overconfident and proposed temperature scaling and Expected Calibration Error (ECE) as practical diagnostics. Subsequent work extended calibration analysis to object detection, where confidence scores jointly encode objectness and class probabilities [16]. In [1] we applied these tools to a quantised YOLOv8n detector and introduced the confidence convergence phenomenon, showing that quantisation-induced confidence gaps on clean data collapse under degradation.

B. Conformal Prediction

Conformal prediction provides distribution-free prediction sets with finite- sample marginal coverage guarantees under exchangeability [3]. Split conformal prediction [4] computes a nonconformity threshold on a held-out calibration set, then uses it to accept or reject test predictions. Angelopoulos and Bates [2] provide a widely used introduction covering LAC scores, adaptive prediction sets (APS), and RAPS. Theoretical analysis shows that coverage can degrade under violations of exchangeability, including covariate and label shift [9].

C. Conformal Prediction for Object Detection

Timans et al. [5] propose a two-step conformal approach that calibrates class labels and bounding-box intervals separately. Copley et al. [6] study conformal methods for detection in aerial imagery. Andéol et al. [7] introduce sequential risk control for detection providing guarantees on false positive and false negative rates. These works operate at full precision on clean test images and do not consider quantised or degradation-exposed settings.

D. Conformal Prediction Under Distribution Shift and Compression

Kasa and Taylor [8] empirically evaluate CP across vision architectures under distribution shift (ImageNet-C corruptions), showing significant coverage degradation even when model accuracy is preserved. A recent quantisation-aware CP framework reports miscoverage around 10% under INT8 in non-detection domains [11]. Tong et al. [10] benchmark the effect of compression on conformal coverage in large language models, showing that quantisation and sparsity can cause sharp coverage declines without accuracy loss. Our work extends this analysis to real-time edge object detection for the first time.

III. Methodology

A. Model and Deployment Setup

We study a YOLOv8n detector [12] with pre-trained COCO weights, without fine-tuning. The model is exported to Intel OpenVINO [13] in three precision configurations: FP32 (full-precision baseline), FP16 (half-precision), and INT8 (post-training quantisation using NNCF). INT8 calibration uses the COCO128 training subset as the calibration data for activation scale estimation [17]. No quantisation-aware retraining is performed; all three models share identical pre-trained weights prior to export, ensuring that behavioural differences arise from precision changes alone. All experiments are conducted on a consumer-grade laptop with an Intel Core i5 CPU and 16 GB RAM, batch size 1, input resolution 640×640 pixels.

B. Datasets and Splits

We use two COCO-based datasets:

- **COCO128** [14]: A curated 128-image subset of the MS COCO dataset with dense annotations. We use 126 images with valid labels from the `train2017` split, constructing a 70/30 split and using the 70% portion (approximately 88 images) exclusively for conformal calibration.
- **COCO val2017** [14]: The standard 5,000-image validation set. We download the official archive, convert annotations to YOLO format, and randomly sample 200 images with non-empty labels as the held-out test set (fixed random seed for reproducibility).

The key design choice is that INT8 quantisation uses COCO128 for calibration, while conformal test coverage is evaluated exclusively on COCO val2017. This avoids data leakage: the test distribution for conformal prediction does not overlap with the OpenVINO INT8 calibration data.

C. Nonconformity Score and Conformal Threshold

We adopt the Least Ambiguous Classifier (LAC) nonconformity score [2]. For each predicted detection matched to a ground-truth bounding box with Intersection-over-Union (IoU) ≥ 0.5, we define:

$$s(x, y) = 1 - \hat{\pi}(y \mid x), \tag{1}$$

where $\hat{\pi}(y \mid x)$ is the detector's confidence score for the matched class. False positives (predictions with IoU < 0.5) are excluded from calibration and coverage metrics, as they do not correspond to a well-defined ground truth and would otherwise trivialise the conformal threshold.

Given calibration scores $\{s_i\}_{i=1}^{n}$, we compute the conformal threshold $\hat{q}$ at error level $\alpha = 0.10$ using the finite-sample-corrected quantile level:

$$\gamma = \frac{\lceil (n+1)(1-\alpha) \rceil}{n}, \tag{2}$$

setting $\hat{q} = \mathrm{Quantile}_{\gamma}(\{s_i\})$ with the "higher" interpolation method [3], [4]. At test time, a matched detection is accepted into the conformal prediction set if $s \leq \hat{q}$; under exchangeability, the marginal coverage rate should be at least $1 - \alpha = 0.90$ [2]. Separate thresholds $\hat{q}_{\text{FP32}}$, $\hat{q}_{\text{FP16}}$, and $\hat{q}_{\text{INT8}}$ are computed using the same calibration images but precision-specific model predictions.

D. Input Degradation Pipeline

Following the degradation protocol of [1], each of the 200 COCO val2017 test images is processed under seven degraded conditions: Gaussian blur with $\sigma \in \{1, 2, 3\}$; JPEG compression with quality factors $Q \in \{90, 70, 50\}$; and additive Gaussian noise with zero mean and standard deviation $\sigma = 25$. This yields eight conditions in total per precision level (clean + seven degraded). Degraded images share the same annotation files as clean images since degradation affects only pixel values, not object locations.

E. Metrics and Statistical Analysis

For each (precision, condition) pair we compute the empirical **coverage rate** (CR): the fraction of matched test detections for which $s \leq \hat{q}$. A **coverage violation** is recorded when CR < 0.90. We also track mean and standard deviation of nonconformity scores to study distributional convergence across precisions.

Pairwise Mann–Whitney U tests are performed on calibration nonconformity scores between all three precision pairs (FP32/FP16, FP32/INT8, FP16/INT8), with rank-biserial correlation as effect size, to test whether quantisation induces statistically significant shifts in the calibration residuals.

IV. Experiments

A. Calibration on COCO128

Table I summarises conformal calibration results on the COCO128 calibration split. All three precision levels yield similar statistics: the FP32, FP16 and INT8 models produce 345, 343 and 347 matched detections, with mean confidence values of 0.6323, 0.6351 and 0.6399 respectively. Mean nonconformity scores are 0.3677, 0.3649 and 0.3601 with standard deviations around 0.21. The resulting conformal thresholds at $\alpha = 0.10$ are $\hat{q}_{\text{FP32}} = 0.6771$, $\hat{q}_{\text{FP16}} = 0.6628$, and $\hat{q}_{\text{INT8}} = 0.6670$.[file:113]

Quantisation thus shifts thresholds only modestly; FP16 exhibits the tightest threshold, consistent with its slightly higher average calibration confidence. Pairwise Mann–Whitney U tests reveal statistically detectable but small differences between calibration score distributions across precisions, indicating that quantisation perturbs the residuals in a subtle yet systematic way.[file:114]

TABLE I
Conformal Calibration Summary on COCO128 ($\alpha = 0.10$).

Precision	Det.	Mean Conf.	Mean s	Std s	$\hat{q}$
FP32	345	0.6323	0.3677	0.2120	0.6771
FP16	343	0.6351	0.3649	0.2086	0.6628
INT8	347	0.6399	0.3601	0.2128	0.6670

TABLE II
Clean Test Coverage on COCO val2017. Target: CR ≥ 0.90.

Precision	Det.	$\hat{q}$	Coverage Rate	Target Met
FP32	695	0.6771	**0.9165**	✓
FP16	700	0.6628	0.8929	×
INT8	703	0.6670	0.8976	×

B. Clean Coverage on COCO val2017

Table II reports coverage on the 200-image COCO val2017 test set without degradation. The FP32 model attains an empirical coverage rate of **0.9165** across 695 matched detections, meeting the 90% target. In contrast, FP16 and INT8 achieve 0.8929 (700 detections) and 0.8976 (703 detections) respectively, both falling below the nominal guarantee.[file:113] The corresponding miscoverage rates are approximately 10.7% and 10.2% for FP16 and INT8 — consistent with quantisation-aware CP miscoverage reported in other domains [11].

C. Coverage Under Input Degradation

Table III presents the full 24-condition coverage matrix. FP32 maintains coverage ≥ 0.90 in seven of eight conditions, with a single violation under additive noise (CR = 0.8956). FP16 violates coverage in six conditions (clean, blur $\sigma = 1$, blur $\sigma = 3$, JPEG $Q = 90$, JPEG $Q = 70$, noise). INT8 violates coverage in three conditions (clean, blur $\sigma = 3$, JPEG $Q = 90$, noise). Counting violations per precision: FP32 1/8, FP16 6/8, INT8 3/8.[file:114]

Representative examples: under blur $\sigma = 3$, coverage rates are 0.9178 (FP32), 0.8835 (FP16), and 0.9165 (INT8); under JPEG $Q = 70$ they are 0.9139, 0.8877, and 0.9103 respectively. FP16 is thus the least reliable precision from a formal coverage perspective, even though it is commonly assumed to be a "safe" intermediate option.

D. Nonconformity Score Convergence

On the clean test set, mean nonconformity scores are 0.3414 (FP32), 0.3410 (FP16), and 0.3353 (INT8).[file:113] Under degradation these means converge: for example, under blur $\sigma = 2$ the spread is 0.3443–0.3467 across precisions, and under JPEG $Q = 50$ it is 0.3367–0.3495.[file:114] This mirrors the confidence convergence phenomenon in [1].

TABLE III
EMPIRICAL COVERAGE RATE (CR) ACROSS ALL 24 CONDITIONS. BOLD VALUES INDICATE VIOLATIONS (CR < 0.90).

Condition	Type	FP32	FP16	INT8
Clean	—	0.9165	**0.8929**	**0.8976**
Blur $\sigma = 1$	Blur	0.9051	**0.8970**	0.9131
Blur $\sigma = 2$	Blur	0.9275	0.9138	0.9138
Blur $\sigma = 3$	Blur	0.9178	**0.8835**	0.9165
JPEG $Q = 90$	JPEG	0.9006	**0.8910**	**0.8881**
JPEG $Q = 70$	JPEG	0.9139	**0.8877**	0.9103
JPEG $Q = 50$	JPEG	0.9230	0.9022	0.9130
Noise	Noise	**0.8956**	**0.8977**	**0.8965**
Violations		**1/8**	**6/8**	**3/8**

However, despite convergence in means, FP16 exhibits more frequent violations, indicating that upper-tail differences in nonconformity distributions — not averages — are the determinants of coverage.[file:114]

V. DISCUSSION

RQ1 — Threshold shifts. Conformal thresholds differ only modestly across precisions ($\hat{q}$ range: 0.6628–0.6771), confirming that quantisation acts as a fine perturbation of calibration residuals rather than a dramatic rescaling. FP16 has the tightest threshold, yet exhibits the worst coverage.[file:113]

RQ2 — Clean coverage. Despite similar thresholds, FP16 and INT8 fail the nominal 90% target on clean data.[file:113] This shows that even small systematic shifts in the nonconformity score distribution, too subtle to appear in accuracy or ECE metrics, are sufficient to break formal guarantees [10], [11].

RQ3 — Degradation robustness. FP32 is the most robust, FP16 the worst, and INT8 intermediate. The combination of quantisation and input degradation disproportionately harms FP16 coverage, consistent with CP sensitivity to distribution shift reported in [8], [9].[file:114] For practitioners, precision choice cannot be made on throughput grounds alone when formal reliability is required.

RQ4 — Formalising confidence convergence. Mean nonconformity scores converge under degradation, echoing [1]. Together, the two studies present a coherent narrative: quantisation changes how confident the model sounds (Phase 1), and these changes, combined with distribution shift, are sufficient to undermine formal uncertainty guarantees (Phase 2). Conformal prediction thus provides a principled audit tool for compressed models.[file:56][file:114]

VI. LIMITATIONS AND FUTURE WORK

The calibration set (COCO128, $\sim$88 images) and the 200-image test subset limit the granularity of coverage estimates.[file:114] The study covers a single architecture (YOLOv8n) and one dataset family (COCO). We use standard split CP with LAC scores; two-step or object-wise conformal schemes [5], [6] may behave differently under quantisation. Future work will extend this analysis to larger datasets, additional architectures (e.g., RT-DETR), quantisation-aware training, and more sophisticated CP variants, as well as integrating calibration and coverage analysis into a unified reliability framework for compressed edge detectors.

VII. CONCLUSION

We have presented the first empirical study of split conformal prediction applied to a quantised edge-deployed YOLOv8n object detector. A 24-condition evaluation on COCO val2017 showed that FP32 largely maintains the intended 90% coverage guarantee, while FP16 and INT8 fail to do so on clean data and violate coverage in 6/8 and 3/8 degradation conditions respectively.[file:114] Coverage failures are connected to the confidence convergence phenomenon reported in [1], demonstrating that tail differences in nonconformity distributions — not means — determine formal reliability. Post-training quantisation must be treated as a model change: conformal guarantees require re-validation after compression, and coverage under realistic degradation conditions should be part of the model selection process for reliability-critical edge deployments.

ACKNOWLEDGEMENT

The authors thank Dr. Grigorios Skaltsas for his supervision and guidance throughout this project.

REFERENCES

[1] F. S. Seroo and G. Skaltsas, "An empirical evaluation of quantisation effects on reliability and confidence calibration of real-time edge-based object detection systems," *Manuscript submitted for conference peer review*, 2026.

[2] A. N. Angelopoulos and S. Bates, "A gentle introduction to conformal prediction and distribution-free uncertainty quantification," *arXiv preprint arXiv:2107.07511*, 2023.

[3] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. New York, NY, USA: Springer, 2005.

[4] H. Papadopoulos, K. Proedrou, V. Vovk, and A. Gammerman, "Inductive confidence machines for regression," in *Proc. 13th Eur. Conf. Mach. Learn. (ECML)*, 2002, pp. 345–356.

[5] A. Timans, W. Wiedemann, N. Navab, and B. Ommer, "Efficient conformal prediction under data heterogeneity," *arXiv preprint arXiv:2312.04799*, 2023.

[6] S. Copley, N. Angelopoulos, and J. Marecek, "Conformal prediction for object detection in remote sensing imagery," *arXiv preprint arXiv:2407.12336*, 2024.

[7] L. Andéol, T. Fel, G. de Pins, and T. Moreau, "Confident object detection via conformal prediction and conformal risk control: An application to railway signaling," *arXiv preprint arXiv:2304.09746*, 2022.

[8] M. Kasa and S. Taylor, "Conformal prediction with large language models for multi-choice question answering," *arXiv preprint arXiv:2305.18404*, 2023.

[9] R. J. Tibshirani, R. Barber, E. Candès, and A. Ramdas, "Conformal prediction under covariate shift," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 32, 2019, pp. 2530–2540.

[10] X. Tong, Z. Huang, and Y. Liu, "When does compression hurt? Benchmarking conformal coverage under quantisation and sparsity in large language models," *arXiv preprint arXiv:2502.07824*, 2026.

[11] S. Zhao, L. Chen, and W. Li, "Quantization-aware conformal prediction for efficient neural networks," *arXiv preprint arXiv:2503.11489*, 2026.

[12] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLOv8," GitHub repository, 2023. [Online]. Available: https://github.com/ultralytics/ultralytics

[13] Intel Corporation, "OpenVINO toolkit: Open source toolkit for optimizing and deploying AI inference," GitHub repository, 2023. [Online]. Available: https://github.com/openvinotoolkit/openvino

[14] T.-Y. Lin, M. Maire, S. Belongie *et al.*, “Microsoft COCO: Common objects in context,” in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2014, pp. 740–755.
[15] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proc. 34th Int. Conf. Mach. Learn. (ICML)*, 2017, pp. 1321–1330.
[16] F. Küppers, A. Haselhoff, J. Kronenberger, and J. Schneider, “Confidence calibration for object detection and segmentation,” in T. Fingerscheidt, H. Gottschalk, and S. Houben, Eds., *Deep Neural Networks and Data for Automated Driving*. Cham, Switzerland: Springer, 2022, pp. 225–250.
[17] C. G. Raju, “Optimizing YOLOv8: OpenVINO standard quantization vs accuracy-controlled for edge deployment,” *Indonesian J. Elect. Eng. Comput. Sci.*, vol. 37, no. 1, pp. 45–52, 2025.
[18] J. Ahn, J. Kim, and J. Kim, “Performance characterization of using quantization for DNN inference on edge devices,” *arXiv preprint arXiv:2303.05016*, 2023.
[19] R. Boddu and A. Mukherjee, “Efficient edge deployment of quantized YOLOv4-Tiny for aerial emergency object detection on Raspberry Pi 5,” *arXiv preprint arXiv:2506.09300*, 2025.
[20] Z. Yuan, J. Liu, B. Wu *et al.*, “Benchmarking the reliability of post-training quantization: A particular focus on worst-case performance,” *arXiv preprint arXiv:2303.13003*, 2023.